ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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17.7IO Structural Checklist for Building Type S4: Dual Systems with Backup Steel Moment Frames and Stiff Diaphragms

Table 17-15. Immediate Occupancy Structural Checklist for Building Type S4.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by gravity loads in frame columns subjected to overturning forces is less than $0.10F_y$. Alternatively, the axial stress caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30F_y$.	5.5.2.1.3	A.3.1.3.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BRACE AXIAL STRESS CHECK: The axial stress in the diagonal braces, calculated using the Quick Check procedure of Section 4.4.3.4 and neglecting the steel moment frame, is less than $0.50F_y$.	5.5.4.1	A.3.3.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE BEARING WALLS: Floor and roof girders and trusses are not supported at the ends of concrete walls that are less than 10 in. (254 mm) thick. This statement only applies to framing supports located less than two times the wall thickness away from the wall end.	5.5.2.5.1	A.3.1.6.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the concrete shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 and neglecting the steel moment frame, is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f'_c}$	5.5.3.1.1	A.3.2.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REINFORCING STEEL: The ratio of shear wall reinforcing steel area to gross concrete area is not less than 0.0012 in the vertical direction and 0.0020 in the horizontal direction. The spacing of reinforcing steel is equal to or less than 18 in. (457 mm).	5.5.3.1.3	A.3.2.2.2	

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Connections						
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad. The anchor rods are capable of resisting the overturning force using the Quick Check procedure of Section 4.4.3.6.	5.7.3.1	A.5.3.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of seismic forces to the shear walls.	5.7.2	A.5.2.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FOUNDATION DOWELS: Wall reinforcement is doweled into the foundation, and the dowels are able to develop the lesser of the strength of the walls or the uplift capacity of the foundation.	5.7.3.4	A.5.3.5

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DRIFT CHECK: The drift ratio of the steel moment frames acting alone, calculated using the Quick Check procedure of Section 4.4.3.1 using 25% of V_c, is less than 0.015.	5.5.2.1.2	A.3.1.3.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of braced frames or shear walls in each principal direction is greater than or equal to 2. The number of braced bays in each line is greater than 3.	5.5.1.1	A.3.2.1.1 A.3.1.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INTERFERING WALLS: All concrete and masonry infill walls placed in moment frames are isolated from structural elements.	5.5.2.1.1	A.3.1.2.1	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel frames, and the connections are able to develop the lesser of the strength of the frames or the diaphragms.	5.7.2	A.5.2.2	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the strength of the adjoining members based on the specified minimum yield stress of the steel.	5.5.2.2.1	A.3.1.3.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PANEL ZONES: All panel zones have the shear capacity to resist the shear demand required to develop 0.8 times the sum of the flexural strengths of the girders framing in at the face of the column.	5.5.2.2.3	A.3.1.3.5	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN SPLICES: All column splice details located in moment frames include connection of both flanges and the web, and the splice develops the strength of the column.	5.5.2.2.4	A.3.1.3.6	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRONG COLUMN–WEAK BEAM: The percentage of strong column–weak beam joints in each story of each line of moment frames is greater than 50%.	5.5.2.1.5	A.3.1.3.7	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BEAM PENETRATIONS: All openings in frame-beam webs are less than one-quarter of the beam depth and are located in the center half of the beams.	5.5.2.2.6	A.3.1.3.9	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	GIRDER FLANGE CONTINUITY PLATES: There are girder flange continuity plates at all moment-resisting frame joints.	5.5.2.2.7	A.3.1.3.10	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OUT-OF-PLANE BRACING: Beam–column joints are braced out of plane.	5.5.2.2.8	A.3.1.3.11	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BOTTOM FLANGE BRACING: The bottom flanges of beams are braced out of plane.	5.5.2.2.9	A.3.1.3.12	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COMPACT MEMBERS: All brace elements meet section requirements in accordance with AISC 360, Table B4.1a.	5.5.4.3	A.3.3.1.7	

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C	NC	N/A	U			
☐	☐	☐	☐	COLUMN SPLICES: All column splice details located in braced frames develop 100% of the tensile strength of the column.	5.5.4.2	A.3.3.1.3
☐	☐	☐	☐	SLENDERNESS OF DIAGONALS: All diagonal elements required to carry compression have *Kl/r* ratios less than 200.	5.5.4.3	A.3.3.1.4
☐	☐	☐	☐	CONNECTION STRENGTH: All the brace connections develop the buckling capacity of the diagonals.	5.5.4.4	A.3.3.1.5
☐	☐	☐	☐	OUT-OF-PLANE BRACING: Braced frame connections that are attached to beam bottom flanges located away from beam–column joints are braced out of plane at the bottom flange of the beams.	5.5.4.5	A.3.3.1.6
☐	☐	☐	☐	K-BRACING: The bracing system does not include K-braced bays.	5.5.4.6	A.3.3.2.1
☐	☐	☐	☐	TENSION-ONLY BRACES: Tension-only braces do not compose more than 70% of the total seismic-force-resisting capacity in structures more than two stories high.	5.5.4.7	A.3.3.2.2
☐	☐	☐	☐	CHEVRON BRACING: Beams in chevron, or V-braced, bays are capable of resisting the vertical load resulting from the simultaneous yielding and buckling of the brace pairs.	5.5.4.6	A.3.3.2.3
☐	☐	☐	☐	CONCENTRICALLY BRACED FRAME JOINTS: All the diagonal braces frame into the beam–column joints concentrically.	5.5.4.8	A.3.3.2.4
☐	☐	☐	☐	COUPLING BEAMS: Coupling beams have the capacity in shear to develop the uplift capacity of the adjacent wall or to develop the flexural capacity of the coupling beam, whichever is less. This statement only applies to coupling beams with span-to-depth ratios exceeding 2-to-1.	5.5.3.2.1	A.3.2.2.3
☐	☐	☐	☐	OVERTURNING: All shear walls have aspect ratios less than 4-to-1. Wall piers need not be considered.	5.5.3.1.4	A.3.2.2.4

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C	NC	N/A	U			
☐	☐	☐	☐	CONFINEMENT REINFORCING: For shear walls with aspect ratios greater than 2-to-1, the boundary elements are confined with spirals or ties with spacing less than $8d_b$.	5.5.3.2.2	A.3.2.2.5
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	WALL REINFORCING AT OPENINGS: There is added trim reinforcement around all wall openings with a dimension greater than three times the thickness of the wall.	5.5.3.1.5	A.3.2.2.6
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	WALL THICKNESS: Thicknesses of bearing walls are not less than 1/25 the unsupported height or length, whichever is shorter, nor less than 4 in. (101.6 mm).	5.5.3.1.2	A.3.2.2.7
Diaphragms (Stiff or Flexible)						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 15% of the wall length.	5.6.1.3	A.4.1.4
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	OPENINGS AT FRAMES: Diaphragm openings immediately adjacent to the braced frames or moment frames extend less than 15% of the frame length.	5.6.1.3	A.4.1.5
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1
Foundation System						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1

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High Seismicity (Complete the Following Items in Addition to the Items for Very Low, Low, and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the strength of the adjoining members or panel zones based on 110% of the expected yield stress of the steel per AISC 341, Section A3.2.	5.5.2.2.1	A.3.1.3.4.	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COMPACT MEMBERS: All moment and braced frame columns and beams meet section requirements in accordance with AISC 341, Table D1.1a, for highly ductile members. Braced frame beams meet section requirements for moderately ductile members.	5.5.2.2.5 5.5.4.3	A.3.3.1.8	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONNECTION STRENGTH: All the brace connections develop the yield capacity of the diagonals.	5.5.4.4	A.3.3.1.5	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation, and the anchorage is able to develop the least of the following: the tensile capacity of the column, the tensile capacity of the lowest-level column splice (if any), or the uplift capacity of the foundation.	5.7.3.1	A.5.3.1	

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